SmartLearn: C-Based Learning & Quiz System Update



1 Project Overview and Core Objectives

Project Introduction



Smartlearn is a C-based, console-oriented educational application that helps students practice questions, take quizzes, and evaluate there academic performance. It includes subject and topic selection, four difficulty choices- easy, medium,hard, and mixed-and automatic score calculation.

Problem Statement & Objectives

Addressing Learning Gaps

SmartLearn is a console-based C application featuring student registration/login, subject- and topic-based questions

Enhancing Practice & Quizzing

practice mode with feedback, quizzes, scoring, result history, performance analysis, and question management.

Improving Performance Tracking

The program runs, but previous quiz results do not display as expected, and personal performance tracking needs improvement.





2 System Features and Technical Implementation

Main Features & Workflow



Student Registration/Login

SmartLearn is a console-based C application featuring student registration/login, subject- and topic-based questions

Practice , Quiz and Add a question Modes

practice mode with feedback, quizzes, scoring, result history, performance analysis, and question management.

Result History & Analysis

scoring, result history, performance analysis, and question management. These are ongoing debugging tasks.

C-Based Technical Implementation

1

Core C Programming Concepts

The project uses C programming concepts such as functions, structures, arrays, loops, conditionals, randomization

2

Input Validation & File Handling

input validation, and file handling. Data files include questions.txt, students.txt, results.txt, and topicstats.txt.

3

Data File Management

Data files include questions.txt, students.txt, results.txt, and topicstats.txt. These are crucial for data persistence.



3 Team Contributions and Project Status

Individual Team Contributions



Abdur Rahman Jabir's Role

Main C program structure and menu, Student registration and login, Question bank and file handling

Ragib Sindid Priam's Role

running the program with various inputs, determining the results and composing a note about testing.

Md. Atikur Rahman's Role

project documentation, including README development, progress tracking, and maintaining clear and updated project information on GitHub.

1

```
Welcome to SmartLearn - practice, quiz, and improve!

=== Student Login ===
Enter your Student ID:
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2

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=== Student Login ===
Enter your Student ID: Labib123
New student. Let's create your profile.
Enter your name: Talha Labib
Profile created. Welcome, Talha Labib!

============================================================
                        SMARTLEARN
   Student: Talha Labib (Labib123)   Questions in bank: 24
============================================================
  1. Practice Mode
  2. Quiz Mode
  3. Performance Analysis (strong / weak topics)
  4. View Past Results
  5. Add a Question to the Bank
  6. Switch Student
  7. Exit
Enter your choice:
```

2

```
Welcome to SmartLearn - practice, quiz, and improve!

=== Student Login ===
Enter your Student ID: Ragib123
Welcome back, Ragib Priam!

============================================================
                        SMARTLEARN
   Student: Ragib Priam (Ragib123)   Questions in bank: 24
============================================================
  1. Practice Mode
  2. Quiz Mode
  3. Performance Analysis (strong / weak topics)
  4. View Past Results
  5. Add a Question to the Bank
  6. Switch Student
  7. Exit
Enter your choice:
```

3

```
Enter your choice: 1

=== Practice Mode ===

Subjects:
  1. Math
  2. Computer Science
Choose subject: 2

Topics in Computer Science:
  1. C Basics
  2. Data Structures
  3. All topics
Choose topic: 3

Difficulty:
  1. Easy
  2. Medium
  3. Hard
  4. Mixed
Choose difficulty: 2
```

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Enter your choice: 2

=== Quiz Mode ===

Subjects:
  1. Math
  2. Computer Science
Choose subject: 1

Topics in Math:
  1. Algebra
  2. Geometry
  3. All topics
Choose topic: 3

Difficulty:
  1. Easy
  2. Medium
  3. Hard
  4. Mixed
Choose difficulty: 3

There are 4 matching questions.
How many questions do you want in the quiz? 4
```

5

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------------------------------------------------------------
Question 1/4   [Algebra | Math]
The sum of the roots of 2x^2 - 7x + 3 = 0 is:
  A) 7/2
  B) -7/2
  C) 3/2
  D) -3/2
Your answer (A-D, S=skip): D

------------------------------------------------------------
Question 2/4   [Geometry | Math]
Number of diagonals in a polygon with 8 sides:
  A) 16
  B) 20
  C) 24
  D) 28
Your answer (A-D, S=skip): A
```

6

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------------------------------------------------------------
QUIZ RESULT - Ragib Priam (Ragib123)
------------------------------------------------------------
Subject    : Math
Topic      : All topics
Difficulty : Hard
Score      : 1 / 4
Percentage : 25.00%
Remark     : Needs improvement. Review the weak topics.

Topic breakdown:
  Algebra              1/2 (50%)
  Geometry             0/2 (0%)

Result saved.
```

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Enter your choice: 5

=== Add a New Question ===
Subject: English
Topic: Right form of verbs
Difficulty (1=Easy, 2=Medium, 3=Hard): 2
Question text: If he _____ harder, he would have passed the exam.
Option A: studied
Option B: had studied
Option C: has studied
Option D: studies
Correct option (1=A, 2=B, 3=C, 4=D): 2
Question added successfully. Total questions: 25
```

8

```
=== Practice Mode ===

Subjects:
  1. Math
  2. Computer Science
  3. English
Choose subject:
```

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```
=== Quiz Mode ===

Subjects:
  1. Math
  2. Computer Science
  3. English
Choose subject:
```



Thank You

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